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Roll No

EE-7004(4)/EX-7004(3)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

SCADA Systems and Applications

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1.
 - a) Define PLC and explain how it is useful in automated process.
 - b) State and explain advantages and disadvantages of PLC in detail.
2.
 - a) Draw and explain the main block diagram of PLC.
 - b) Explain the rules for constructing ladder diagram.
3. Explain the following:
 - i) Intelligent electronics devices
 - ii) SCADA server
4. Draw and explain SCADA architecture in detail along with its applications.
5. Explain the following:
 - i) Open systems interconnection protocols
 - ii) Control net protocol

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6.
 - a) Write a short note on energy management system.
 - b) Discuss the different system operating states.
7. Write short notes on SCADA system used in
 - i) Petroleum refining
 - ii) Water purification system
8. Explain the following (Any two)
 - a) Interfacing of PLC with SCADA
 - b) Remote terminal unit
 - c) System security and state estimation
 - d) SCADA applications
